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Department of Electronics & Communication Engineering

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

MCP PBL Report

“Wireless Vehicle-to-Vehicle (V2V) Communication System Using UART and HC-12 433 SI4463 Wireless Serial Module”

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Introduction

Vehicle-to-Vehicle (V2V) communication is emerging as a critical layer in collision-avoidance systems, complementing traditional sensor-based approaches such as ultrasonic, radar, and camera-based detection. While onboard sensors are effective for line-of-sight obstacle detection, they fail in scenarios involving occluded approaches — such as vehicles converging at a T-junction or blind intersection — where no direct sensor line exists between the two vehicles until it is too late to react safely.

This project implements a two-vehicle V2V collision-avoidance system using STM32F446RE microcontrollers, where each vehicle broadcasts its own pose (position and heading) over a low-cost HC-12 wireless radio link. By exchanging this pose data, each vehicle computes a real-time Time-to-Collision (TTC) estimate against the other, enabling proactive GO/DEVIATE/STOP decisions regardless of approach angle. The system further incorporates a baseline finite state machine (FSM) as a safety-verified decision layer, with a trained spiking neural network (SNN) as an optional, more adaptive decision-making alternative. A local HC-SR04 ultrasonic sensor serves as a fallback emergency stop mechanism in the event of radio link loss, ensuring the system degrades gracefully rather than failing silently.

Problem Statement

The proposed project aims to address the following challenges associated with vehicle safety and communication:

- Drivers often have limited reaction time to respond to sudden braking or unexpected obstacles, increasing the risk of road accidents.
- Conventional vehicles operate independently and cannot exchange real-time safety information with nearby vehicles.
- Most existing Vehicle-to-Vehicle (V2V) communication technologies, such as DSRC and C-V2X, require expensive infrastructure and complex communication protocols.
- Low-cost vehicles and educational prototype systems require a simpler and more economical wireless communication solution.
- There is a need for a system that can continuously monitor vehicle conditions, such as obstacle distance and sudden deceleration, and transmit this information wirelessly with minimal delay.
- The system should provide timely warning alerts to nearby vehicles, enabling drivers to take preventive action and reduce the possibility of collisions.

To address these challenges, this project develops a **wireless Vehicle-to-Vehicle communication system** using **STM32F407VG microcontrollers**, **HC-12 (SI4463) 433 MHz Wireless Serial Modules**, **HC-SR04 ultrasonic sensors**, and **MPU6050 accelerometers**. The proposed system enables the real-time transmission of safety-related information between vehicles using **UART communication**, improving driver awareness while providing a cost-effective and reliable solution for collision avoidance.

Objectives

The primary objectives of the proposed project are:

- To design and implement a low-cost V2V wireless communication link between two STM32-based vehicle platforms using HC-12 radio modules for real-time pose broadcasting.
- To compute Time-to-Collision (TTC) from exchanged pose data (position and heading) as a means of detecting collision risk independent of physical line-of-sight between vehicles.
- To implement a deterministic finite state machine (FSM) as a baseline GO/DEVIATE/STOP decision policy, ensuring predictable and verifiable safety behavior.
- To develop and evaluate a spiking neural network (SNN) as an alternative, adaptive decision-making layer trained on logged sensor and TTC data, while retaining the FSM as a ground-truth safety reference.
- To implement a local fallback mechanism using an HC-SR04 ultrasonic sensor and last-known-trajectory estimation, ensuring safe degradation in the event of radio link failure.
- To validate the system's effectiveness in a non-line-of-sight T-junction scenario, demonstrating collision risk detection that a single-vehicle sensor-only system could not achieve.

Methodology

The proposed Vehicle-to-Vehicle (V2V) communication system is developed using two STM32F407VG microcontrollers representing the transmitting and receiving vehicles. The overall methodology of the system is divided into the following stages:

1. System Initializations

- Initialize the STM32F407VG microcontroller and configure all required peripherals.
- Configure UART for communication with the HC-12 wireless module.
- Configure the I²C interface for the MPU6050 sensor and OLED display.
- Configure GPIO pins and timers required for the HC-SR04 ultrasonic sensor.

2. Sensor Data Acquisition

- The HC-SR04 ultrasonic sensor continuously measures the distance between the vehicle and nearby obstacles.
- The MPU6050 sensor continuously monitors the vehicle's acceleration and deceleration along the X, Y, and Z axes.
- The STM32F407VG reads the sensor values at regular intervals and converts them into usable digital data.

3. Data Processing and Packet Formation

- The acquired sensor readings are processed by the STM32F407VG microcontroller.
- The processed values are organized into a structured UART data packet containing:
 - Obstacle distance
 - Acceleration values (X, Y, and Z axes)
 - Brake/Alert status
- A packet header and delimiter are included to ensure accurate data transmission and reception.

4. Wireless Data Transmission

- The formatted data packet is transmitted through the HC-12 (SI4463) 433 MHz Wireless Serial Module using UART communication.
- The HC-12 module wirelessly sends the packet to the receiver vehicle without requiring additional communication infrastructure.

5. Wireless Data Reception

- The HC-12 module on the receiver vehicle receives the transmitted packet.
- The receiver STM32F407VG reads the incoming UART data and extracts the transmitted sensor information.
- The received packet is verified to ensure correct communication before further processing.

6. Decision Making

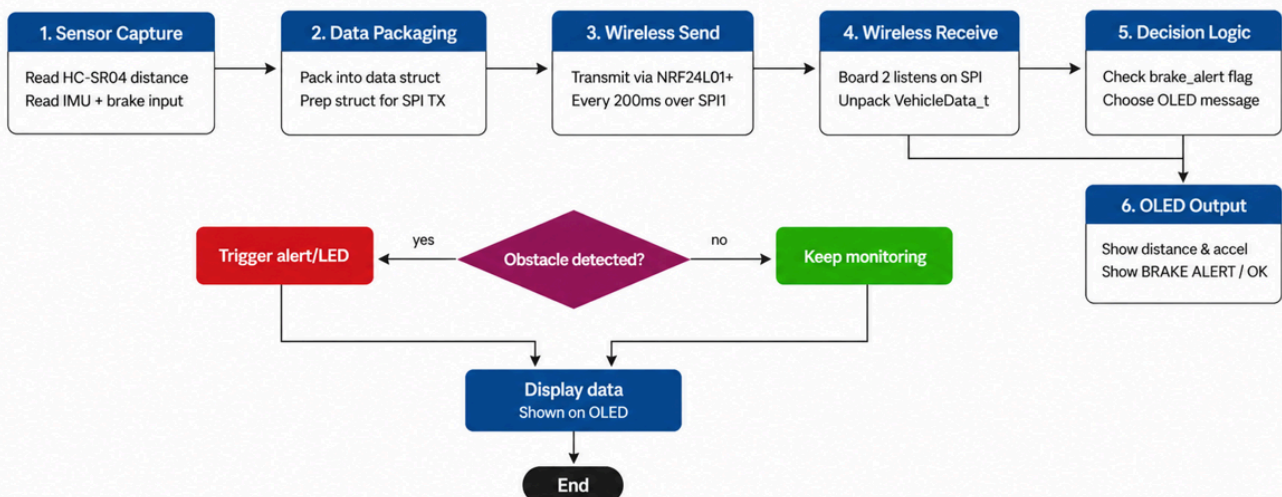
- The receiver compares the received distance and acceleration values with predefined safety thresholds.
- If the obstacle distance is below the safe limit or sudden braking is detected, the system identifies a potential collision risk.
- Otherwise, the system continues monitoring the incoming sensor data.

7. Alert Generation and Display

- Under normal driving conditions, the OLED displays real-time information such as obstacle distance, acceleration values, and system status.
- If a hazardous condition is detected, warning messages such as **"Obstacle Detected"** or **"Brake Alert"** are displayed on the OLED.
- A visual alert is also generated using an LED to notify the driver immediately.

8. Continuous Monitoring

- The complete sensing, processing, wireless transmission, reception, decision-making, and alert generation process is repeated continuously.
- This enables real-time Vehicle-to-Vehicle communication and ensures timely warning messages are delivered whenever unsafe driving conditions occur.



Hardware Tools / Platforms

- **STM32F446RE Nucleo boards ×2**
Cortex-M4 microcontroller boards used as the main embedded control platforms.
- **HC-12 433 MHz UART Radio Module**
Used for the V2V wireless broadcast communication link.
- **HC-SR04 Ultrasonic Sensor**
Used for local fallback obstacle detection.
- **MPU-6050 IMU**
Used for Gyro-Z based heading estimation.
- **DRV8833 Dual H-Bridge Motor Driver**
Used for DC motor control.
- **N20 Geared DC Motors with Quadrature Encoders**
Used for vehicle movement and speed/position feedback.

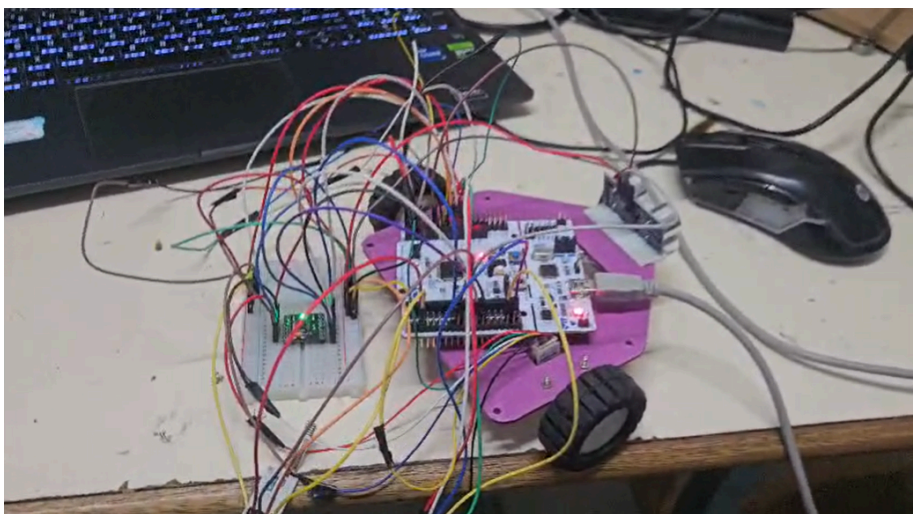
Software / Development Tools

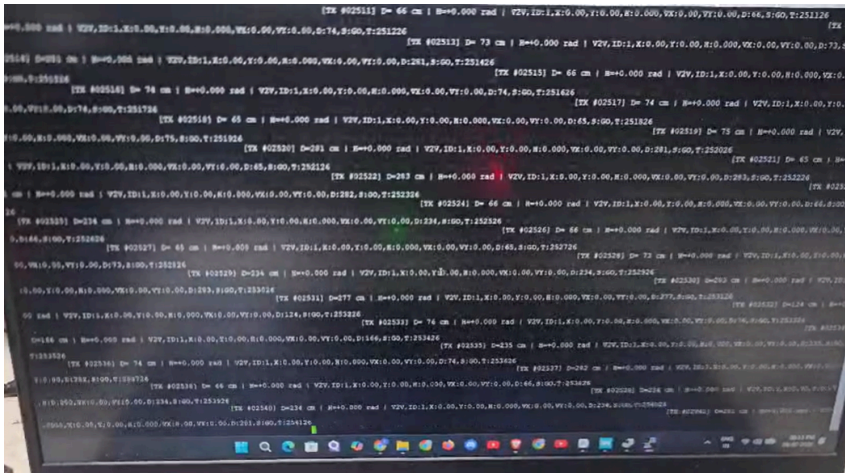
- **STM32CubeIDE + STM32CubeMX**
Used for peripheral configuration, code generation, development, and debugging.
- **STM32 HAL Library**
Used for hardware abstraction and peripheral-level programming.
- **Python**
Used for testing, data logging, visualization, and supporting scripts.

Results & Conclusions

This project demonstrates a low-cost V2V communication system using STM32 microcontrollers and HC-12 wireless modules. Real-time data from the ultrasonic sensor and MPU-6050 is transmitted between two vehicle nodes and processed to detect possible collision risks.

The system provides timely safety alerts through an OLED display, showing that basic wireless V2V communication can be implemented effectively using simple embedded hardware. It also creates a base for future improvements such as GPS integration, multi-vehicle communication, and automatic braking support.





Important Instructions

1. Report should not be more than 4 pages
2. Please follow the font type of Times New Roman with size 14 for the headings and size 12 for text.
3. The complete document should be justified (distribute the text evenly between the margins)
4. The document should be printed on both sides of a paper.
5. Do not copy paste the PPT slides directly here.

Status:	Accepted/Not Accepted

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